

Problem 8: Coprime Representation of Graphs

A detailed solution

24 May 2026

Abstract

The problem asks for the largest integer size needed to represent an arbitrary n -vertex graph by an injective labelling of its vertices with positive integers, where two vertices are adjacent exactly when their labels are coprime. The question proposed the order $2^{\Theta(n)}$. This note gives a detailed proof that the proposed order is false. In fact,

$$R(n) = \exp(\Theta(n \log n)) = 2^{\Theta(n \log n)} = n^{\Theta(n)}.$$

The lower bound already follows from the graph obtained by taking a clique on $n - 1$ vertices and adding one isolated vertex: the isolated vertex must share a different prime factor with each clique vertex, forcing a primorial-sized label. The upper bound is the standard nonedge-prime construction.

1 Problem statement

For a positive integer M , write

$$[M] = \{1, 2, \dots, M\}.$$

For positive integers a, b , write (a, b) for their greatest common divisor. All graphs in this note are finite, simple, and undirected.

Problem 8 (Coprime representation of graphs). *For a finite graph G , let $\rho(G)$ be the least M such that the vertices of G can be injectively labelled by integers in $[M]$ with*

$$uv \in E(G) \iff (\ell(u), \ell(v)) = 1.$$

Let

$$R(n) = \max_{|V(G)|=n} \rho(G).$$

Is

$$R(n) = 2^{\Theta(n)}?$$

The answer is no. The correct order is larger by a factor of $\log n$ in the exponent.

2 Main theorem

Let p_i denote the i -th prime, so $p_1 = 2$, $p_2 = 3$, $p_3 = 5$, and so on.

Theorem 1. *There is an absolute constant $C > 0$ such that, for every $n \geq 3$,*

$$\prod_{i=1}^{n-1} p_i \leq R(n) \leq (Cn^2 \log n)^n.$$

Consequently

$$R(n) = \exp(\Theta(n \log n)) = 2^{\Theta(n \log n)} = n^{\Theta(n)}.$$

In particular, $R(n)$ is not $2^{\Theta(n)}$.

The proof is split into the lower bound and the upper bound. The lower bound is the main new observation: a single isolated vertex attached to a large clique in the complement forces one label to have many distinct prime divisors.

3 The lower bound

For $n \geq 3$, let

$$G_n = K_{n-1} \sqcup \{z\}.$$

Thus G_n has vertices

$$v_1, v_2, \dots, v_{n-1}, z,$$

where $v_1, \dots, v_{n-1}$ span a clique, and z is isolated. Equivalently,

$$v_i v_j \in E(G_n) \quad (i \neq j), \quad z v_i \notin E(G_n) \quad (1 \leq i \leq n-1).$$

Proposition 1. *For $n \geq 3$,*

$$\rho(G_n) = \prod_{i=1}^{n-1} p_i.$$

Proof. We prove the lower and upper inequalities separately.

Step 1: every valid labelling forces a primorial-sized label

Let

$$\ell : V(G_n) \rightarrow [M]$$

be any valid injective labelling satisfying

$$xy \in E(G_n) \iff (\ell(x), \ell(y)) = 1.$$

Since z is not adjacent to any v_i , the defining condition gives

$$(\ell(z), \ell(v_i)) > 1 \quad \text{for every } 1 \leq i \leq n-1.$$

Therefore, for each i , there is at least one prime divisor common to $\ell(z)$ and $\ell(v_i)$. Choose one such prime and call it q_i . Thus

$$q_i \mid \ell(z) \quad \text{and} \quad q_i \mid \ell(v_i).$$

We now show that the primes

$$q_1, q_2, \dots, q_{n-1}$$

must be pairwise distinct. Suppose not. Then for some distinct indices i, j we would have

$$q_i = q_j = q.$$

By construction, q_i divides $\ell(v_i)$ and q_j divides $\ell(v_j)$. Hence the same prime q divides both $\ell(v_i)$ and $\ell(v_j)$, so

$$(\ell(v_i), \ell(v_j)) \geq q > 1.$$

But v_i and v_j are adjacent because the vertices $v_1, \dots, v_{n-1}$ form a clique. The labelling rule therefore requires

$$(\ell(v_i), \ell(v_j)) = 1,$$

a contradiction. Hence the primes $q_1, \dots, q_{n-1}$ are indeed pairwise distinct.

Since all q_i divide $\ell(z)$ and they are distinct primes, their product also divides $\ell(z)$:

$$q_1 q_2 \cdots q_{n-1} \mid \ell(z).$$

Therefore

$$\ell(z) \geq q_1 q_2 \cdots q_{n-1}.$$

Among all products of $n - 1$ distinct primes, the smallest possible product is the product of the first $n - 1$ primes. Indeed, after arranging the distinct primes q_i increasingly as

$$r_1 < r_2 < \cdots < r_{n-1},$$

one has $r_i \geq p_i$ for every i , because p_i is the i -th smallest prime. Thus

$$q_1 q_2 \cdots q_{n-1} = r_1 r_2 \cdots r_{n-1} \geq p_1 p_2 \cdots p_{n-1}.$$

Combining these inequalities gives

$$\ell(z) \geq \prod_{i=1}^{n-1} p_i.$$

Since every label lies in $[M]$, in particular $\ell(z) \leq M$. Hence every valid labelling satisfies

$$M \geq \prod_{i=1}^{n-1} p_i.$$

It follows that

$$\rho(G_n) \geq \prod_{i=1}^{n-1} p_i.$$

Step 2: the primorial bound is attained

Now define a concrete labelling of G_n by

$$\ell(z) = \prod_{i=1}^{n-1} p_i, \quad \ell(v_i) = p_i \quad (1 \leq i \leq n-1).$$

This labelling is injective for $n \geq 3$: the labels $p_1, \dots, p_{n-1}$ are distinct primes, and the label of z is their product, which is larger than each individual prime.

We check the coprimality condition. If $i \neq j$, then

$$(\ell(v_i), \ell(v_j)) = (p_i, p_j) = 1,$$

because two distinct primes are coprime. Thus every pair v_i, v_j is correctly represented as an edge.

On the other hand,

$$(\ell(z), \ell(v_i)) = \left(\prod_{j=1}^{n-1} p_j, p_i \right) = p_i > 1.$$

Thus z is correctly represented as nonadjacent to every v_i . These are all pairs of vertices in G_n , so the labelling is valid. Its largest label is

$$\prod_{i=1}^{n-1} p_i.$$

Therefore

$$\rho(G_n) \leq \prod_{i=1}^{n-1} p_i.$$

Together with the lower inequality from Step 1, this proves

$$\rho(G_n) = \prod_{i=1}^{n-1} p_i.$$

□

Since G_n is an n -vertex graph, Proposition 1 gives immediately

$$R(n) \geq \rho(G_n) = \prod_{i=1}^{n-1} p_i.$$

This is already much larger than 2^{cn} on the exponential scale. To make that explicit, note that

$$p_i \geq i + 1 \quad (i \geq 1),$$

because the i -th prime is at least the $(i + 1)$ -st positive integer. Hence

$$\prod_{i=1}^{n-1} p_i \geq \prod_{i=1}^{n-1} (i + 1) = n!.$$

Taking logarithms and comparing the sum with an integral,

$$\log(n!) = \sum_{j=1}^n \log j \geq \int_1^n \log x \, dx = n \log n - n + 1.$$

Therefore

$$R(n) \geq n! \geq \exp(n \log n - O(n)).$$

This proves the lower bound

$$\log R(n) \geq \Omega(n \log n).$$

4 The upper bound

The upper bound is the natural construction: give every nonedge its own prime, and put that prime into exactly the two endpoint labels. Then two labels share a prime exactly when the corresponding vertices form a nonedge.

We shall use the following elementary supply of small primes. A proof is included in the appendix.

Lemma 1 (Enough small primes). *There is an absolute constant $A > 0$ such that the k -th prime satisfies*

$$p_k \leq Ak \log(k+1) \quad (k \geq 1).$$

Equivalently, for every $k \geq 1$, there are at least k distinct primes not exceeding $Ak \log(k+1)$.

Let G be an arbitrary graph on n vertices. We will construct a valid labelling whose largest label is at most

$$(Cn^2 \log n)^n$$

for an absolute constant C .

Let $\overline{E}(G)$ denote the set of nonedges of G :

$$\overline{E}(G) = \{\{u, v\} : u \neq v, uv \notin E(G)\}.$$

For every nonedge $\{u, v\} \in \overline{E}(G)$, choose a prime p_{uv} . Also, for every vertex $v \in V(G)$, choose a private prime q_v . All these primes are chosen to be distinct from each other.

The total number of primes needed is

$$|\overline{E}(G)| + |V(G)| \leq \binom{n}{2} + n \leq n^2 \quad (n \geq 2).$$

By Lemma 1, we may choose all these primes among the first n^2 primes, and hence each chosen prime is at most

$$An^2 \log(n^2 + 1) \leq Cn^2 \log n$$

for a suitable absolute constant C and all $n \geq 3$.

Define the label of a vertex v by

$$\ell(v) = q_v \prod_{\substack{u \in V(G) \\ u \neq v, uv \notin E(G)}} p_{uv}.$$

In words, $\ell(v)$ contains the private prime q_v , and for each nonneighbor u of v , it also contains the nonedge-prime p_{uv} .

We check three things: injectivity, correctness on nonedges, and correctness on edges.

Injectivity

Let $v \neq w$. The private prime q_v divides $\ell(v)$. By construction, q_v appears in no other vertex label: it is not any nonedge-prime p_{xy} , and it is not the private prime of another vertex. Thus

$$q_v \nmid \ell(w).$$

If $\ell(v)$ and $\ell(w)$ were equal, then every prime dividing $\ell(v)$ would also divide $\ell(w)$, including q_v . This is impossible. Hence

$$\ell(v) \neq \ell(w).$$

So the labelling is injective.

Correctness on nonedges

Suppose $uv \notin E(G)$. Then the prime p_{uv} was assigned to the nonedge uv . By definition of the labels, p_{uv} divides both $\ell(u)$ and $\ell(v)$. Therefore

$$(\ell(u), \ell(v)) \geq p_{uv} > 1.$$

Thus nonadjacent vertices are represented by noncoprime labels.

Correctness on edges

Suppose now that $uv \in E(G)$. We claim that no prime can divide both $\ell(u)$ and $\ell(v)$.

Indeed, every prime used in the construction is of exactly one of the following two types.

- A private prime q_x . Such a prime divides only the label of the single vertex x .
- A nonedge-prime p_{xy} . Such a prime divides exactly the labels of the two endpoints x and y of the nonedge xy .

A private prime cannot divide both $\ell(u)$ and $\ell(v)$ because $u \neq v$. A nonedge-prime p_{xy} can divide both $\ell(u)$ and $\ell(v)$ only if its two endpoints are exactly u and v , that is, only if $\{x, y\} = \{u, v\}$. But a prime p_{uv} was assigned only when uv was a nonedge. Here uv is an edge, so no such nonedge-prime exists.

Therefore $\ell(u)$ and $\ell(v)$ have no common prime factor, and hence

$$(\ell(u), \ell(v)) = 1.$$

Thus adjacent vertices are represented by coprime labels.

Size of the labels

Each label $\ell(v)$ is a product of at most n primes: one private prime plus at most $n - 1$ nonedge-primes, one for each possible nonneighbor. Each chosen prime is at most $Cn^2 \log n$. Hence

$$\ell(v) \leq (Cn^2 \log n)^n \quad \text{for every } v \in V(G).$$

Thus

$$\rho(G) \leq (Cn^2 \log n)^n.$$

Since G was an arbitrary n -vertex graph, taking the maximum over all such G gives

$$R(n) \leq (Cn^2 \log n)^n.$$

5 Conclusion

Combining the lower and upper bounds, for $n \geq 3$ we have

$$\prod_{i=1}^{n-1} p_i \leq R(n) \leq (Cn^2 \log n)^n.$$

The lower bound satisfies

$$\log \left(\prod_{i=1}^{n-1} p_i \right) \geq \log(n!) \geq n \log n - O(n),$$

and the upper bound satisfies

$$\log((Cn^2 \log n)^n) = n \log(Cn^2 \log n) = O(n \log n).$$

Therefore

$$\log R(n) = \Theta(n \log n),$$

which is equivalent to

$$R(n) = \exp(\Theta(n \log n)) = 2^{\Theta(n \log n)} = n^{\Theta(n)}.$$

This disproves the proposed estimate $R(n) = 2^{\Theta(n)}$, since $2^{\Theta(n)} = \exp(\Theta(n))$ has only linear, not $n \log n$, growth in the exponent.

A A self-contained proof of the small-primes lemma

We prove Lemma 1. This is a standard Chebyshev-type argument, included to justify that the nonedge-prime construction can be carried out using primes of polynomial size.

For $x \geq 1$, define

$$\psi(x) = \sum_{p^a \leq x} \log p,$$

where the sum is over all prime powers $p^a \leq x$, with $a \geq 1$. Equivalently, each prime $p \leq x$ contributes

$$\left\lfloor \frac{\log x}{\log p} \right\rfloor \log p$$

to $\psi(x)$.

Lemma 2. *There is an absolute constant $a > 0$ such that, for all sufficiently large x ,*

$$\psi(x) \geq ax.$$

Proof. Let $m \geq 1$ be an integer. We first show that

$$\psi(2m) \geq \log \binom{2m}{m}.$$

Fix a prime p . By Legendre's formula, the exponent of p in the central binomial coefficient is

$$\nu_p \binom{2m}{m} = \sum_{j \geq 1} \left(\left\lfloor \frac{2m}{p^j} \right\rfloor - 2 \left\lfloor \frac{m}{p^j} \right\rfloor \right).$$

For each j , the summand is either 0 or 1, because

$$\lfloor 2y \rfloor - 2\lfloor y \rfloor \in \{0, 1\}$$

for every real number y . Also, the summand is 0 whenever $p^j > 2m$. Therefore

$$\nu_p \binom{2m}{m} \leq \#\{j \geq 1 : p^j \leq 2m\} = \left\lfloor \frac{\log(2m)}{\log p} \right\rfloor.$$

This means that the prime-power product encoded by $\psi(2m)$ contains every prime factor of $\binom{2m}{m}$ with at least the required multiplicity. Hence

$$\binom{2m}{m} \leq \exp(\psi(2m)),$$

and so

$$\psi(2m) \geq \log \binom{2m}{m}.$$

Next, the central binomial coefficient is the largest term in the expansion

$$\sum_{r=0}^{2m} \binom{2m}{r} = 2^{2m} = 4^m.$$

There are $2m + 1$ terms, so the largest term is at least the average term:

$$\binom{2m}{m} \geq \frac{4^m}{2m + 1}.$$

Therefore

$$\psi(2m) \geq \log \binom{2m}{m} \geq m \log 4 - \log(2m + 1).$$

Now let x be large and choose $m = \lfloor x/2 \rfloor$. Then $2m \leq x$, so $\psi(x) \geq \psi(2m)$. Also $2m \geq x - 1$. Thus

$$\psi(x) \geq (x - 1) \log 2 - \log(x + 1).$$

For all sufficiently large x , the right-hand side is at least $(\log 2/3)x$. Hence the lemma holds, for example with $a = \log 2/3$ after increasing the threshold for x if necessary. $\square$

Let $\pi(x)$ be the number of primes at most x . Since each prime $p \leq x$ contributes at most $\log x$ to $\psi(x)$, we have

$$\psi(x) = \sum_{p \leq x} \left\lfloor \frac{\log x}{\log p} \right\rfloor \log p \leq \sum_{p \leq x} \log x = \pi(x) \log x.$$

Combining this with Lemma 2, we get a Chebyshev-type lower bound

$$\pi(x) \geq \frac{ax}{\log x}$$

for all sufficiently large x .

We now deduce the bound on the k -th prime. Choose a fixed constant $B > 0$ so large that, for every $k \geq 2$,

$$\frac{aB \log(k + 1)}{\log(Bk \log(k + 1))} \geq 1.$$

Such a B exists because

$$\log(Bk \log(k+1)) = O_B(\log(k+1))$$

uniformly for $k \geq 2$, while the numerator is $aB \log(k+1)$. Set

$$x = Bk \log(k+1).$$

For B also chosen large enough to put x beyond the threshold in the Chebyshev-type bound, we have

$$\pi(x) \geq \frac{aBk \log(k+1)}{\log(Bk \log(k+1))} \geq k.$$

Thus the k -th prime is at most $Bk \log(k+1)$ for every $k \geq 2$. Increasing B once more, if necessary, also covers $k = 1$. This proves

$$p_k \leq Ak \log(k+1)$$

for an absolute constant A , as required.